

Yu Xia

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EDUCATION

University of California, Riverside

Master of Science in Robotics

Sep. 2023 – Jun. 2025

Riverside, CA, USA

Wuhan Huaxia University of Technology

Bachelor of Engineering in Computer Science and Technology

Sep. 2017 – Jun. 2021

Wuhan, China

PUBLICATIONS

BOLT: Bandwidth-Efficient Object-Level Streaming for XR Robot Teleoperation

SenSys 2027 (Under Review)

Jun. 2026

- **Yu Xia**, Hang Qiu
- Designed an object-level XR streaming framework for robot teleoperation that transmits object 6D poses and dynamically generated meshes instead of full 3D scenes under strict motion-to-photon latency budgets.
- Built a full-stack Meta Quest prototype, reducing uplink traffic to **0.4 Mbps** ($\sim 220\times$ lower than point cloud streaming) and improving noisy-pose translation error from **67.0** to **25.1 mm** via cross-view refinement.

Towards Generalizable Safety in Crowd Navigation via Conformal Uncertainty Handling

CoRL 2025

Sep. 2024

- Jianpeng Yao, Xiaopan Zhang*, **Yu Xia***, Zejing Wang, Amit K. Roy-Chowdhury, Jiachen Li
- Designed a conformal inference-guided RL framework for crowd navigation, achieving **96.93% success rate** and **3.72 $\times$ fewer collisions** with robust out-of-distribution performance.
- Deployed the system on TurtleBot 4 for real-world crowd interaction in both sparse and dense environments.

NL2SpaTiaL: Translating Ambiguous Natural Language into Spatio-Temporal Logic Specifications

IROS 2026 (Under Review)

Nov. 2025

- Licheng Luo, Kaier Liang, **Yu Xia**, Mingyu Cai
- Proposed a geometry-aware synthesis pipeline integrating qualitative spatial reasoning with correctness-preserving back-translation for generalizable NL $\rightarrow$ SpaTiaL translation.

ROBOT HARDWARE EXPERIENCE

TurtleBot 4 — Differential-Drive Mobile Robot

- Differential-drive mobile robot with iRobot Create 3 base, Raspberry Pi 4, OAK-D stereo camera, and RPLIDAR A1M8 2D LiDAR.
- Deployed as the primary sim-to-real platform for the conformal inference-guided crowd navigation policy (CoRL 2025). LiDAR-based pedestrian detection provides 3D positions as input to the RL policy for real-time avoidance in dynamic indoor environments.

Yahboom ROSMASTER X3 — Omnidirectional Mobile Robot

- Mecanum-wheel omnidirectional mobile robot with Jetson Orin, SLAMTEC LiDAR, and Astra depth camera.
- Used as an additional sim-to-real platform for the crowd navigation work (CoRL 2025), validating policy generalization across different mobile robot morphologies with the same LiDAR-based pedestrian detection pipeline.

ALOHA 2 — Bimanual Teleoperation System

- Low-cost bimanual teleoperation system with 4 $\times$ Trossen ViperX/WidowX 6-DoF arms, leader-follower control, and multi-view RGB cameras.
- Served as the real-robot evaluation platform for NL2SpaTiaL (IROS 2026). Action trajectories generated by Pi0 are evaluated against spatial STL formulas to verify whether executed manipulations satisfy the translated spatio-temporal specifications.

Kinova Gen3 — 7-DoF Collaborative Arm

- 7-DoF ultra-lightweight collaborative arm (8.2 kg, 902 mm reach, 4 kg payload) with integrated torque sensors and 2D/3D vision module.

- Serving as the manipulation execution end for an ongoing VR teleoperation project. Integrating Kortex API with WebRTC-based remote control for real-time manipulation over WAN.

Hexapod Robot (DIY) — Legged Locomotion Platform

- Custom-built six-legged robot for RL-based locomotion research, including gait training and flip-recovery behavior.
- Trained RL locomotion and fall-recovery policies in simulation. Developed the real-robot controller for deployment; sim-to-real transfer pending due to hardware actuator limitations.

PROJECT EXPERIENCE

TeleopXR — VR Teleoperation for Robot Manipulation over WAN

Jan. 2026 – Present

Advisor: Prof. Hang Qiu, UCR CISL Lab | Ongoing Project

Riverside, CA

- Designed and implemented a VR-based teleoperation system for remote robot manipulation with intuitive hand-tracking control via extended reality interfaces.
- Engineered a hybrid WebRTC + Tailscale architecture to overcome ROS 2 DDS limitations on WAN/5G networks, achieving low-latency real-time communication.
- Developed the full-stack pipeline from VR hand tracking to robot execution, including ROS 2 integration, Docker containerization, and remote development workflows.

Keywords: ROS 2, WebRTC, Tailscale, Meta Quest, Kinova Gen3, Docker, Unity, 5G

Robotics Fabrication: Ball Balancing Car

Apr. 2024

- **Role:** Perception & Control Developer. Built subsystems for sensing, wireless communication, and PID motor control for real-time ball balancing.

Keywords: Arduino, PID, IMU, Bluetooth

UAV Path Planning and Obstacle Avoidance

Dec. 2023

- **Role:** Simulator & Communication Developer. Set up simulator, configured UAV sensors, established ROS–PX4 communication for RL training.
- Developed a rule-based baseline for navigating obstacles; contributed to RL reward function design to accelerate convergence.

Keywords: ROS, PX4, Gazebo, PPO

Mask Wearing Detection with YOLOv5

Sep. 2021

- Trained and optimized YOLOv5 for face mask verification, achieving **85% accuracy** on SF-MASK via iterative hyperparameter tuning.

Keywords: YOLOv5, PyTorch, OpenCV

Image Classification System Design

May 2021

- Enhanced AlexNet architecture and achieved **85% accuracy** on CIFAR-10 through comparative experiments using TensorFlow.

Keywords: TensorFlow, AlexNet, CIFAR-10

SKILLS

Programming: Python, C/C++, Java, MATLAB, HTML5, SQL

Robotics & Sim: ROS / ROS 2, MoveIt 2, Gazebo, MuJoCo, Isaac Sim, PX4, Unity

AI / ML: PyTorch, TensorFlow, OpenCV, MediaPipe, YOLO, Reinforcement Learning

Tools: Docker, Git, Linux (Ubuntu), WebRTC, SolidWorks, Blender, Photoshop

Robot Platforms: TurtleBot 4, ROSMASTER X3, ALOHA 2, Kinova Gen3, Hexapod (DIY), UAV (PX4)

SUMMARY

Robotics M.S. from UC Riverside with work at CoRL 2025 and papers under review at IROS 2026/SenSys 2027. Hands-on experience in robot manipulation, deep reinforcement learning, XR teleoperation, and sim-to-real deployment across 6 physical robot platforms. Proficient in Python, C++, ROS 2, MoveIt 2, PyTorch, and simulation tools (Gazebo, MuJoCo, Isaac Sim). U.S. Permanent Resident.